MSG No. 10.47 P_a2/m [Type IV, monoclinic]

Table 1: Wyckoff site: `2a`, site symmetry: `2/m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[5,6,7,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `2/m`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,7,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `2/m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8]`

Table 4: Wyckoff site: `2d`, site symmetry: `2'/m`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, 0, 0\right]$	`[2,3,5,8]`

Table 5: Wyckoff site: `2e`, site symmetry: `2'/m`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,3,5,8]`

Table 6: Wyckoff site: `2f`, site symmetry: `2/m`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8]`

Table 7: Wyckoff site: `2g`, site symmetry: `2'/m`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,4,6,7]`

2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[2,3,5,8]`

Table 8: Wyckoff site: `2h`, site symmetry: `2'/m`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,5,8]`

Table 9: Wyckoff site: `4i`, site symmetry: `2`

No.	position	mapping
1	$[0, y, 0]$	`[1,2]`
2	$[0, -y, 0]$	`[3,4]`
3	$\left[\frac{1}{2}, y, 0\right]$	`[5,6]`
4	$\left[\frac{1}{2}, -y, 0\right]$	`[7,8]`

Table 10: Wyckoff site: `4j`, site symmetry: `2'`

No.	position	mapping
1	$\left[\frac{1}{4}, y, 0\right]$	`[1,6]`
2	$\left[\frac{3}{4}, y, 0\right]$	`[2,5]`
3	$\left[\frac{3}{4}, -y, 0\right]$	`[3,8]`
4	$\left[\frac{1}{4}, -y, 0\right]$	`[4,7]`

Table 11: Wyckoff site: `4k`, site symmetry: `2`

No.	position	mapping
1	$\left[0, y, \frac{1}{2}\right]$	`[1,2]`
2	$\left[0, -y, \frac{1}{2}\right]$	`[3,4]`
3	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[5,6]`
4	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[7,8]`

Table 12: Wyckoff site: `4l`, site symmetry: `2'`

No.	position	mapping
1	$\left[\frac{1}{4}, y, \frac{1}{2}\right]$	`[1,6]`
2	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	`[2,5]`
3	$\left[\frac{3}{4}, -y, \frac{1}{2}\right]$	`[3,8]`
4	$\left[\frac{1}{4}, -y, \frac{1}{2}\right]$	`[4,7]`

Table 13: Wyckoff site: `4m`, site symmetry: `m`

No.	position	mapping
1	$[x,\, 0,\, z]$	`[1,4]`
2	$[-x,\, 0,\, -z]$	`[2,3]`
3	$\left[x+\frac{1}{2},\, 0,\, z\right]$	`[5,8]`
4	$\left[\frac{1}{2}-x,\, 0,\, -z\right]$	`[6,7]`

Table 14: Wyckoff site: `4n`, site symmetry: `m`

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, z\right]$	`[1,4]`
2	$\left[-x,\, \frac{1}{2},\, -z\right]$	`[2,3]`
3	$\left[x+\frac{1}{2},\, \frac{1}{2},\, z\right]$	`[5,8]`
4	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, -z\right]$	`[6,7]`

Table 15: Wyckoff site: `8o`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-x,\, y,\, -z]$	`[2]`
3	$[-x,\, -y,\, -z]$	`[3]`
4	$[x,\, -y,\, z]$	`[4]`
5	$\left[x+\frac{1}{2},\, y,\, z\right]$	`[5]`
6	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	`[6]`
7	$\left[\frac{1}{2}-x,\, -y,\, -z\right]$	`[7]`
8	$\left[x+\frac{1}{2},\, -y,\, z\right]$	`[8]`